NABIEL MUHAMMAD IMJAUZANANSYAH

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PROFESSIONAL SUMMARY

Fresh Information Systems graduate (GPA 3.66/4.00) with full-stack software engineering experience across backend architecture, REST API integration, and database design. Built and shipped production-style systems: a Node.js/Express.js backend with JWT authentication and real-time WebSocket data (InvestMentor), a Laravel-based internal web application with REST API integrations (Telkomsel), and a modular Python ETL pipeline. Comfortable working across the stack - React.js frontends, Node.js/Express and Laravel backends, MySQL/PostgreSQL databases - with additional applied experience integrating machine learning models into production services.

EDUCATION

Bachelor of Information Systems, Universitas Muhammadiyah Sumatera Utara (UMSU)

Medan, Indonesia | 2022 - June 2026

- GPA: 3.66 / 4.00
- Final year project: real-time BISINDO sign language translator with a React.js frontend, Node.js backend, and CNN-LSTM inference service (96% validation accuracy);
- published in Jurnal ICT: Information and Communication Technologies (2026)
- Relevant coursework: Software Engineering, Web Development, Database Systems, Algorithm and Programming, Machine Learning, Data Analysis

RELEVANT EXPERIENCE

Software Developer Intern, PT Telekomunikasi Selular (Telkomsel)

Medan, Indonesia | Jun 2025 - Sep 2025

- Designed and built a Python-based Telegram voucher-distribution bot (Bot Voucher BGA SUMBAGUT) integrated with the internal SAMOSIR.LIVE platform via REST APIs and real-time webhooks, implementing reward redemption, challenge, survey, point management, whitelist, and automated broadcast features
- Engineered end-to-end encryption, two-factor authentication, and fraud-detection safeguards, and architected the system to scale from 100 to 1,000 transactions per hour during peak periods
- Deployed the bot to production in September 2025, achieving 100% completion of the assigned scope (exceeding the initial 80% target) and a 98/100 supervisor evaluation score; the bot has remained in continuous operation for 10+ months since, well beyond the internship period
- Collaborated with internal developers on API integration, testing, debugging, and deployment preparation; enhanced Laravel-based internal web modules and managed data interactions with MySQL

Machine Learning Engineer Trainee, DBS Foundation Coding Camp 2025

Remote, Part-time | Feb 2025 - Aug 2025

- Completed an intensive engineering program covering data preprocessing, model development, evaluation, and deployment workflows
- Built and shipped data-driven prototypes, including a recommendation system and an NLP-based text classification model, using Python, Scikit-learn, TensorFlow, and Keras
- Practiced reproducible project structuring and technical documentation to support production-ready workflows
- Completed the ParentCare capstone project in a 6-member team, delivering an AI-powered parenting platform with 3 full-stack features and 3 machine learning models, integrating predictive models into a production-ready web application.

PROJECTS

InvestMentor - AI-Powered Trading Simulator & Agentic Mentor Platform

Full-Stack Engineer | Independent Project | Sep 2025 - Jun 2026

- Designed and built a Node.js/Express.js backend with a secure REST API architecture, implementing JWT authentication, API rate limiting, and database transactions via Sequelize ORM
- Built real-time data ingestion pipelines integrating the Yahoo Finance API and Google News RSS, streaming dynamic market context to the frontend via WebSocket (Socket.IO)

- Integrated the Google Gemini 2.0 Flash SDK to power a multi-turn conversational AI mentor with context-aware responses for stock market analysis
- Deployed an LSTM-based TensorFlow.js model to generate daily stock price predictions with calculated confidence levels

Fashion Studio ETL Pipeline - Independent Data Engineering Project

Mar 2025

- Built a modular ETL pipeline in Python to extract web data, transform raw product information, validate data types, and load results into CSV, Google Sheets, and PostgreSQL
- Implemented data cleaning and validation for price, rating, colors, size, and gender fields; used unit tests to verify extract, transform, and load functions
- Practiced database connection management and error handling across the pipeline

NzSignify - Real-Time BISINDO Sign Language Translator

Final Year Project | Dec 2025 - Apr 2026

- Built a React.js frontend integrated with a Node.js backend to process real-time webcam streams and communicate with an AI inference service via REST API
- Engineered a hybrid CNN-LSTM deep learning pipeline for gesture classification, achieving 96% validation accuracy on an original 11,000-frame dataset across 11 gesture classes
- Implemented Text-to-Speech (TTS) functionality to convert recognized gestures into spoken output, improving accessibility for end users

Additional project: ParentCare - AI-powered parenting style platform, integrated a Random Forest classifier with a Node.js backend via child-process, returning predictions through a REST API (team of 6, DBS Foundation Capstone, 2025).

TECHNICAL SKILLS

Programming Languages: JavaScript, Node.js, PHP, Python

Backend & Database: Express.js, Laravel, JWT Authentication, Sequelize ORM, MySQL, PostgreSQL, SQL Fundamentals, Relational Database Design, CRUD Operations, Query Writing

Data Engineering & Integration: ETL Pipelines, Real-Time Data Pipelines, REST API Integration, WebSocket (Socket.IO), Google Sheets API, Pandas, NumPy

Applied AI & Machine Learning: TensorFlow, TensorFlow.js, CNN-LSTM, LSTM, LLM Integration (Google Gemini API), Scikit-learn

Tools & Platforms: Git, GitHub, Docker, VS Code, Vercel, Railway, Google Colab

CERTIFICATIONS

- Git and GitHub Fundamentals - Dicoding & Google Developers (2025)
- SQL Fundamentals - Dicoding & Google Developers (2025)
- Fundamentals of Data Processing / ETL Pipelines - Dicoding & Google Developers (2025)
- Telkomsel Official Internship Certificate - Telkomsel BGA Regional SUMBAGUT (2025)
- Additional: DBS Foundation Coding Camp (Machine Learning Engineering), Applied Machine Learning, Machine Learning Development, Machine Learning for Beginners, Data Analysis with Python - Dicoding & Google Developers (2025)